

Zad.1 Wyznacz równanie prostej przechodzącej przez punkty A i B:

a) A (4, 3), B (-1, 4) b) A (6, $-\frac{1}{2}$), B (5, 0) c) A ($-\sqrt{2}$, 4), B (2, $\frac{1}{3}$)

1) Podstawiamy oba punkty do prostej
i rozwiązujemy układ równań:

a) A (4, 3) B (-1, 4)

$$y = ax + b$$

$$A(4, 3) \in y = ax + b \Leftrightarrow 3 = 4a + b$$

$$B(-1, 4) \in y = ax + b \Leftrightarrow 4 = -a + b$$

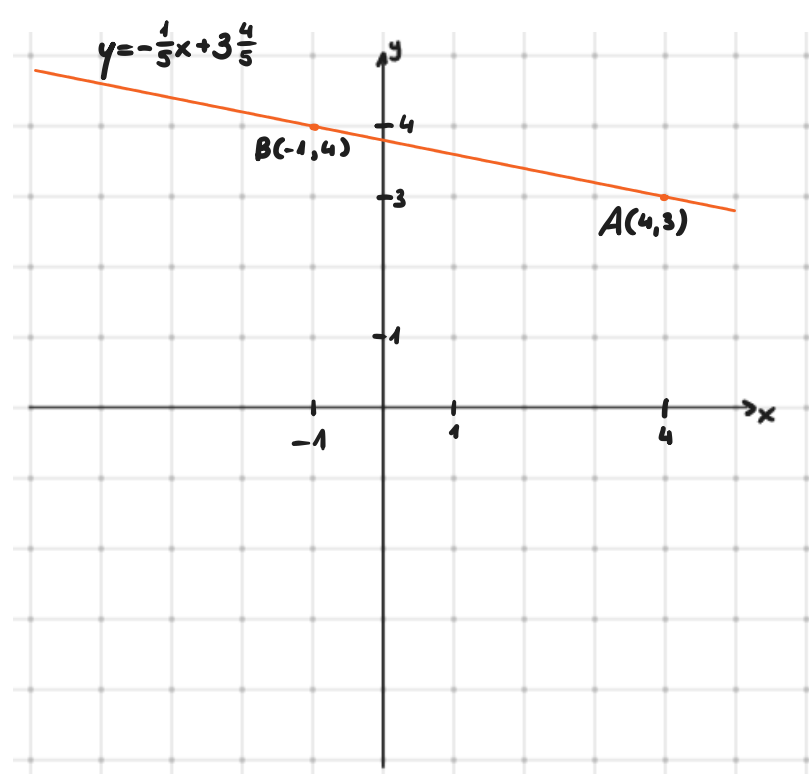
$$\begin{cases} 3 = 4a + b \\ 4 = -a + b \end{cases} \Rightarrow b = 4 + a$$

$$3 = 4a + 4 + a$$

$$-1 = 5a$$

$$a = -\frac{1}{5} \Rightarrow b = 4 + a = 4 - \frac{1}{5} = 3\frac{4}{5}$$

Odp. a) $y = -\frac{1}{5}x + 3\frac{4}{5}$.



b) A (6, $-\frac{1}{2}$) B (5, 0)

$$y = ax + b$$

$$A(6, -\frac{1}{2}) \in y = ax + b \Leftrightarrow -\frac{1}{2} = 6a + b$$

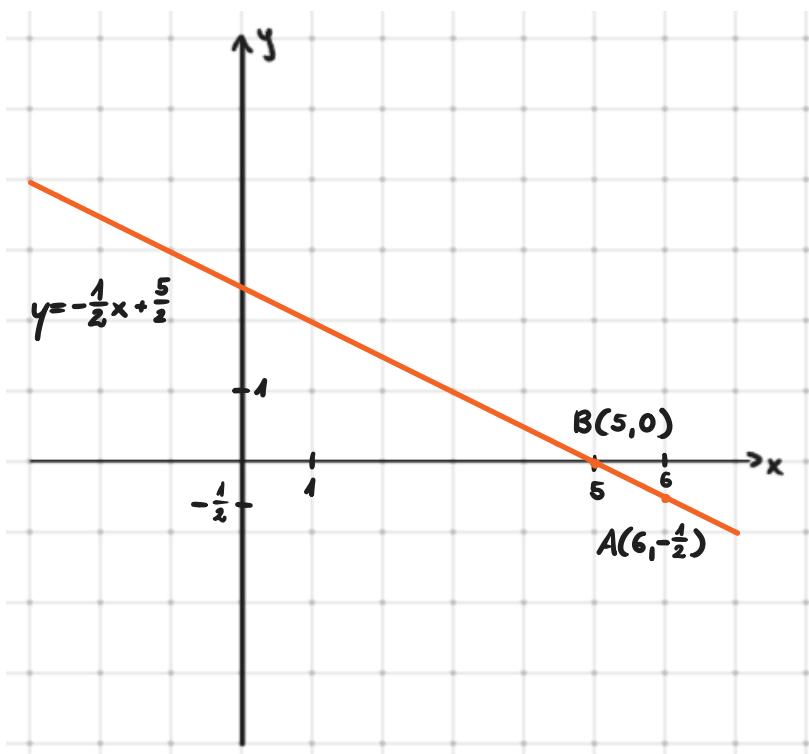
$$B(5, 0) \in y = ax + b \Leftrightarrow 0 = 5a + b$$

$$\begin{cases} -\frac{1}{2} = 6a + b \\ 0 = 5a + b \end{cases} \Rightarrow b = -5a$$

$$-\frac{1}{2} = 6a - 5a$$

$$a = -\frac{1}{2} \Rightarrow b = -5(-\frac{1}{2}) = \frac{5}{2}$$

Odp. b) $y = -\frac{1}{2}x + \frac{5}{2}$.



c) A ($-\sqrt{2}$, 4) B (2, $\frac{1}{3}$)

$$y = ax + b$$

$$A(-\sqrt{2}, 4) \in y = ax + b \Leftrightarrow 4 = -\sqrt{2}a + b$$

$$B(2, \frac{1}{3}) \in y = ax + b \Leftrightarrow \frac{1}{3} = 2a + b$$

$$\begin{cases} 4 = -\sqrt{2}a + b \\ \frac{1}{3} = 2a + b \end{cases} \Rightarrow b = \frac{1}{3} - 2a$$

$$4 = -\sqrt{2}a + \frac{1}{3} - 2a$$

$$\frac{11}{3} = -(2 + \sqrt{2})a$$

$$a = \frac{-11}{3(2 + \sqrt{2})} = \frac{-11(2 - \sqrt{2})}{3(2 + \sqrt{2})(2 - \sqrt{2})} =$$

$$= \frac{11(\sqrt{2} - 2)}{3(4 - 2)} = \frac{11(\sqrt{2} - 2)}{6}$$

$$b = \frac{1}{3} - 2 \cdot \frac{11(\sqrt{2} - 2)}{6} = \frac{1 - 11(\sqrt{2} - 2)}{3}$$

Odp. c) $y = \frac{11(\sqrt{2} - 2)}{6}x + \frac{1 - 11(\sqrt{2} - 2)}{3}$.

